Finite repository rho counts and their accounting

The actual repository benchmark completes768/768 known-answer calls: 576 measured and192 warmups, with no failure or interruption. Every measured deterministic replay matches. The192 measured and64 warmup same-seed A/A pairs have identical counts, phases, native counters and answers. A separately copied-index one-counter corruption is detected. SessionECBS1h99ad02ca65c0. [pcc:unverified]

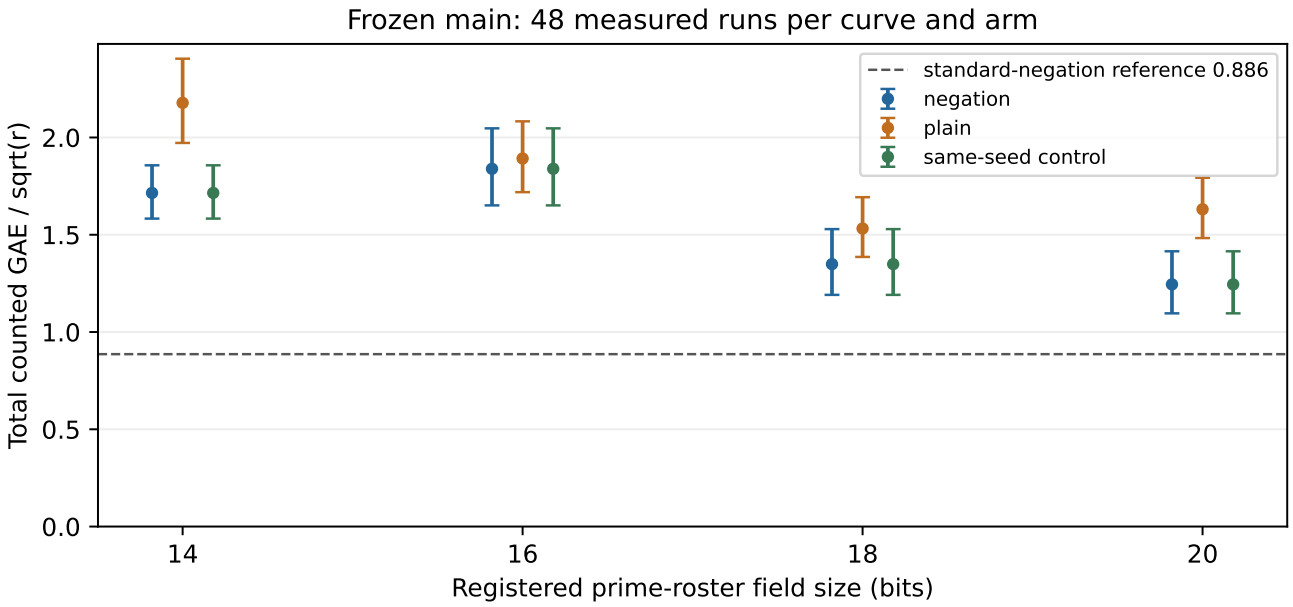

Figure 1: Native recorded means and descriptive95% intervals, rounded as printed by ecbench table.48measured runs percurve/arm:16targets and3seeded rounds. All costs include setup, search and internal verification. The dashed0.886 line is a standard-negation model reference, not a universal lower bound. [pcc:unverified]

All four curves are registered generic prime-roster inputs with automorphism count2 and cofactor1. The A/A arm repeats negation rho on identical seeds; it tests determinism, not independent-seed variability. Main and preflight populations are separate. All scientific clocks are invalid; only counted operations are reported. [pcc:unverified]

The observed means lie above sqrt(pi/4). These small fixed inputs do not refute or establish the qualified asymptotic average. Every size is below the harness's2^25 calibration range, and fixed setup is explicit. Correct source hypotheses and arithmetic cost models remain intact. [pcc:unverified]

Cold totals, emitted counters and honest comparisons

S is counted group-addition equivalents divided by sqrt(r), with an addition and doubling charged1 each and scalar multiplication charged by its actual additions/doublings. The interval covers method setup through recovery and internal verification. All rows use48measured calls percurve/ arm. The A/A arm equals the negation arm in every paired record. [pcc:unverified]

Registered curve	r	Negation S0	Plain S0	Negation S0.1	Negation S1
icv1-fp14-tm43-158b6914	16231	1.715	2.178	1.829	2.857
icv1-fp16-t351-d433f24e	65003	1.839	1.892	1.962	3.075
icv1-fp18-t175-45770ebc	261673	1.349	1.532	1.453	2.381
icv1-fp20-t1293-3c2999ba	1046999	1.245	1.631	1.353	2.328

The band prices every emitted unpriced operation at0/0.1/1GAE. Negation emits canonicalisation counts; plain lists none, so its displayed band is constant. This is the emitted-counter band, not a calibrated complete bit-operation model. The ordering reverses at weight1 on every curve. No ranking holds throughout the declared band. Classification: accounting, since changing the price changes numbers while the algorithms are fixed. No isolated causal effect of negation is claimed: the two registered methods also have their documented walk/cycle policies. [pcc:unverified]

r	Negation setup	Search	Internal check	Total	Emitted unpriced
16231	2669	6935	882	10486	6983
65003	6401	15088	1011	22500	15136
261673	6644	25283	1206	33133	25331
1046999	6744	53123	1293	61160	53171

The r column joins the registered curves above. These are exact integer sums over48measured negation calls. Every plain/A/A phase and counter is retained in analysis/main/native.jsonl and the canonical session. [pcc:unverified]

The separate12-call preflight verifies and replays completely. Its seeds/ targets are distinct; it is not pooled into the main figures. The original adapter export omitted analysis-script metadata and was retained as an unvalidated draft; a new completed pilot record passes. Main metadata was completed before its first export write. No solver outcome was retried. [pcc:unverified]

Formal arithmetic, provenance and scope

The reporting Lean project proves arithmetic on the explicitly recorded totals. It does not prove native execution authenticity, compiler refinement, curve cardinalities or a limiting collision law. The native and formal assurance scopes are separate. Full local gate11/0failing/0trusted-native. [pcc:unverified]

RhoConstantReportAudit declaration	Exact checked obligation
normalized_interval	Exact squared inequalities imply positive-denominator mean bounds
recorded_mean_intervals	Four rational intervals of width0.0001 for the recorded means
exact_phase_totals	Setup + search + internal verification equals each recorded total
emitted_weight_order_flips	Weight0/1 ordering reverses for all four sets of recorded integers
recorded_means_above_reference	Fixed recorded means exceed sqrt(pi/4), with no expectation assertion
finite_report_arithmetic_certificate	Combined finite report arithmetic

Lean4.35.0 / Mathlib56a00585; only propext, Classical.choice and Quot.sound. No holes, custom axioms, native_decide, trusted-native exceptions, waived linters or new elaboration options. Local replay is distinct from independently signed standing. Proof/source/packet receipts accompany the study. [pcc:unverified]

Main protocol045a59cc was frozen and committed01eb80f3 before data; preflight protocol3892337a was frozen and committed1214cbda before its separate data. Main64targets,3rounds,1warmup,3arms. Binary61030c23 is pinned exactly, with3,098 selected source/input files matching baseline 1d2ad3fd. Actual optimized release build succeeded after retained source- coverage errors and connection interruptions. Full hashes are in protocols and evidence/build-provenance.json. [pcc:unverified]

The executable's Git metadata is runtime-derived, not a compile-time revision receipt. Actual main session Git commit is null and is not backfilled. Exact compiled-input and binary hashes remain authoritative. Native SQLite3.46.1 analysis source/binary/header/runtime/query hashes are frozen. Flow768planned, attempted,completed;0notcompleted/interrupted/resumed/excluded. Every scientific clock has wall_validfalse. No independent-machine or signed- execution claim. No paid compute or storage. [pcc:unverified]

Canonical session files remain loose for native replay paths under the13a exception and approved zero-paid local checkpoint. Both exec/ directories are empty. Generated SQLite index is rebuildable derived data, not canonical evidence; no .db is committed. All measurements, warmups, controls and failures are retained. Reproduction commands and exact files: study README. [pcc:unverified]

Primary reading: Bernstein–Lange SafeCurves survey, selected4/4.1–4.5/5intro/ B.1–B.2. It distinguishes batched-affine additions and cost models; sqrt(pi/4) is not claimed universally optimal. Web PDF text succeeded after bounded page/PDF failures; no full-paper read invented. KB unavailable, bounded repository/primary/Mathlib fallback. [pcc:unverified]

This is requirement11's finite benchmark component. Generic p-adic/complexity/ native-refinement, limiting-law proofs and the remaining fourteen-item requirements stay open. No novelty, new security conclusion or refutation of a correctly qualified source statement. [pcc:unverified]